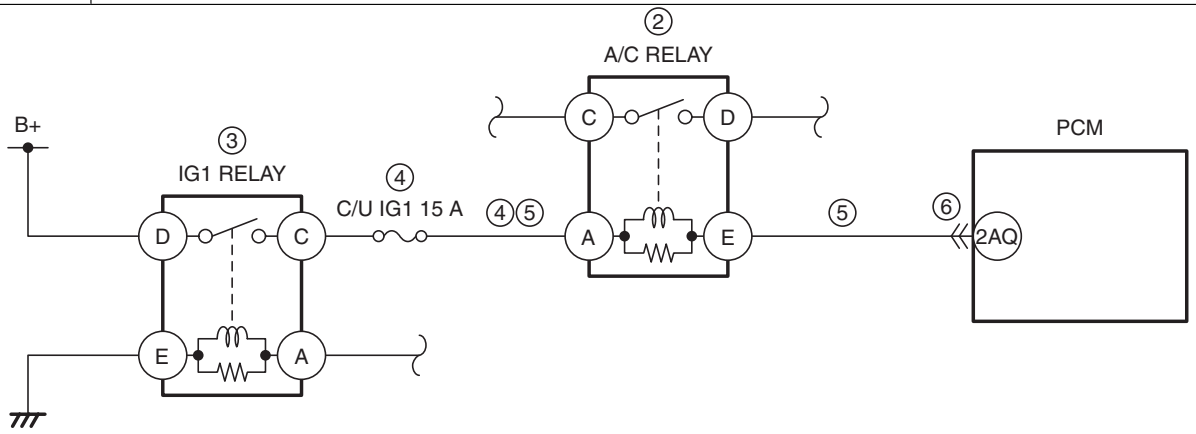
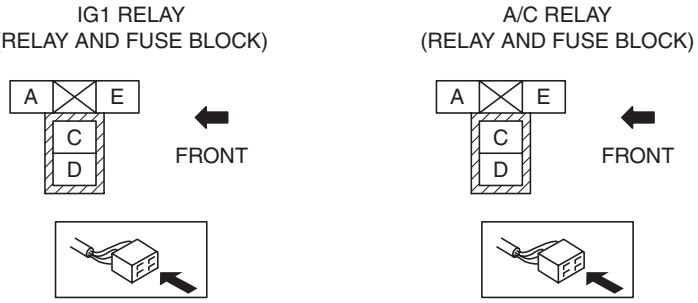
DTC P0646:00 [SKYACTIV-D 1.5]

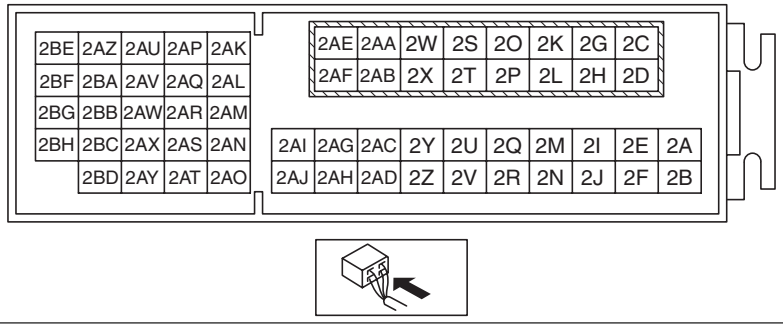
id0102q2587900

DTC P0646:00	A/C relay circuit low input
DETECTION CONDITION	• When the following conditions are met, PCM detects that the A/C relay voltage is **approx. 0 V** for a continuous **5 s**: **MONITORING CONDITIONS** — Battery voltage: **9—16 V** — A/C relay controlled off **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine-transaxle integration control. (ATX)
POSSIBLE CAUSE	• A/C relay malfunction • IG1 relay malfunction • C/U IG1 15 A fuse malfunction • Short to ground in wiring harness between the following terminals: — IG1 relay terminal C—A/C relay terminal A — A/C relay terminal E—PCM terminal 2AQ • PCM connector or terminals malfunction • PCM malfunction





PCM WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**INSPECT A/C RELAY** • Switch the ignition off. • Remove the A/C relay. (See RELAY LOCATION.) • Inspect the A/C relay. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the A/C relay, then go to Step 7. (See RELAY LOCATION.)
		No	Go to the next step.
3	**INSPECT IG1 RELAY** • Remove the IG1 relay. (See RELAY LOCATION.) • Inspect the IG1 relay. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the IG1 relay, then go to Step 7. (See RELAY LOCATION.)
		No	Go to the next step.
4	**INSPECT C/U IG1 15 A FUSE** • Remove the C/U IG1 15 A fuse. (See RELAY LOCATION.) • Inspect the C/U IG1 15 A fuse. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between C/U IG1 15 A fuse and A/C relay terminal A. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the fuse. If the fuse is damaged: • Replace the fuse. Go to Step 7.
		No	Reinstall the C/U IG1 15 A fuse, then go to the next step. (See RELAY LOCATION.)
5	**INSPECT A/C RELAY CIRCUIT FOR SHORT TO GROUND** • Verify that the A/C relay and IG1 relay are removed. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — A/C relay terminal A — A/C relay terminal E • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — IG1 relay terminal C—A/C relay terminal A — A/C relay terminal E—PCM terminal 2AQ **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 7.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to the next step.
		No	Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine. • Turn the air conditioner on and off. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
8	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.